

PRACTICAL NO:5

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ROLL NO:272

INF = 999999

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graph = [  
    [0, 4, 2, 0, 0],  
    [4, 0, 1, 5, 0],  
    [2, 1, 0, 8, 10],  
    [0, 5, 8, 0, 2],  
    [0, 0, 10, 2, 0]  
]  
  
vertices = ["Computer", "IT", "Mechanical", "Library", "Office"]  
V = len(vertices)  
  
# ----- Prim ----- #  
  
def prim():  
    selected = [False] * V  
    selected[0] = True  
    edges = 0  
    cost = 0  
    print("\nMinimum Spanning Tree using Prim\n")  
    while edges < V - 1:  
        minimum = INF  
        x = 0  
        y = 0  
        for i in range(V):  
            if selected[i]:  
                for j in range(V):
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        if (not selected[j]) and graph[i][j]:
            if graph[i][j] < minimum:
                minimum = graph[i][j]
                x = i
                y = j

    print(vertices[x], "--", vertices[y], "=", graph[x][y])
    cost += graph[x][y]
    selected[y] = True
    edges += 1

    print("\nMinimum Cost =", cost)
# ----- Kruskal ----- #

parent = []

def find(i):
    while parent[i] != i:
        i = parent[i]
    return i

def union(i, j):
    a = find(i)
    b = find(j)
    parent[a] = b

def kruskal():
    edges = []
    for i in range(V):
        for j in range(i + 1, V):
            if graph[i][j] != 0:
                edges.append((graph[i][j], i, j))

    edges.sort()

    for i in range(V):
        parent.append(i)

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cost = 0

print("\nMinimum Spanning Tree using Kruskal\n")

count = 0

for w, u, v in edges:
    if find(u) != find(v):
        union(u, v)
        print(vertices[u], "--", vertices[v], "=", w)
        cost += w
        count += 1
        if count == V - 1:
            break

print("\nMinimum Cost =", cost)

# ----- Main Menu ----- #

while True:
    print("\n1. Prim")
    print("2. Kruskal")
    print("3. Exit")
    ch = int(input("Enter Choice : "))
    if ch == 1:
        prim()
    elif ch == 2:
        parent.clear()
        kruskal()
    elif ch == 3:
        break
    else:
        print("Invalid Choice")

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